Indoor Positioning Systems Based on a Modified Matrix-Weighted Fusion Estimator With Multipath and NLOS Mitigation

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***Abstract*—In this article, an indoor positioning system is investigated for an unmanned ground vehicle based on inertial measurement unit (IMU) and ultrawide band (UWB) integration. Measurement outliers of UWB usually exist due to multipath and non-line-of-sight (NLOS) effects. To describe characteristics of stochastic noises well, heavy-tailed noises modeled by Student's t distributions are considered. By a NLOS identification method, a local modified fusion estimator is designed based on the Kalman filter framework to fuse the measurements of IMU and UWB. A modified matrix-weighted fusion estimator is proposed for high-accuracy estimation and robustness. Sufficient conditions are given for the fusion estimation error to be exponentially bounded in mean square. Finally, effectiveness of the proposed fusion estimator is indicated by experimental results.**

***Index Terms*—Fusion estimator, heavy-tailed noises, indoor positioning system, non line of sight (NLOS).**

I. Introduction

HIGHT-ACCURACY indoor positioning technology is of great significance for Internet of Things applications [1], target tracking [2], inventory management [3], and other fields. In recent years, many kinds of wireless indoor positioning technologies are booming, such as Wi-Fi, radio frequency identification (RFID), and ultrawide band (UWB). Based on received signal strength indicators, Wi-Fi-based positioning technology estimates the distance between the terminal and the known location by using a certain channel fading model assumed in the current environment [4]. However, real-time performance and anti-interference are poor because signals transmitted over Wi-Fi are susceptibly disturbed [5]. Although RFID-based positioning technology does not require manual intervention, the lower positioning accuracy is unfortunate [6], [7]. UWB stands out among indoor positioning methods due to its low-cost and low-power consumption [8]. However, any independent localization method has its drawbacks [9]. The positioning accuracy of UWB is mainly affected by multipath effect and non-line-of-sight (NLOS) propagation [10], which is inevitable in practical scenarios.

NLOS identification is necessary for the positioning system to improve positioning accuracy before the measurement information is used to calculate the position [11]. In order to identify whether the measurement is subjected to NLOS interference, machine learning techniques are developed to analyze the real measurements of UWB in different scenarios in [12]. Based on channel information, an identification method with less dependence on the environment is proposed, which is able to determine a specific NLOS channel [13]. Note that UWB provides absolute positioning information for the system, and inertial measurement unit (IMU) can provide relative positioning information with no NLOS effect [14]. However, using IMU alone for dead reckoning positioning, the cumulative error will become larger and larger as the time increases. Naturally, measurement of UWB is usually fused with that of IMU to provide more accurate location estimation [15]. Integrated indoor positioning systems are proposed by combining IMU and UWB based on standard Kalman filter (KF) [16] and unscented KF [17], respectively. In extremely confined environments, the adaptive extended KF is utilized in a positioning system to mitigate the localization performance instability [18]. An adaptive radio-based extended KF is presented in [19] to estimate the measurement noise covariance, which does not require a priori statistical knowledge of the noise. Further, hybrid algorithm is proposed to improve the estimation performance by fusing IMU and UWB.

Note that information fusion can make the best use of effective data from UWB and IMU to obtain high-precision estimated results. However, standard KF, extended KF or unscented KF all assume that process noise and measurement noise follow Gaussian distribution [20], [21]. In practice, due to multipath effect, NLOS and unknown disturbances, the process noise and measurement noise have characteristics of heavy-tailed [22]. In [23], a filter that model heavy-tailed noises as a Student's t distribution is proposed for the system affected by measurement outliers. A distributed fusion estimator is derived for multisensor systems with non-Gaussian but heavy-tailed noises [24]. An event-triggered distributed fusion estimator is designed to deal with heavy-tailed noises in multisensor systems [25]. A sequential fusion estimator is studied for Markov jump systems subjected to heavy-tailed noises with Student's t distribution [26]. However, as far as we know, existing algorithms only consider NLOS

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identification, or only consider heavy-tailed noises in practical applications. Therefore, it is an interesting work to investigate fusion estimates for positioning systems by simultaneously considering NLOS identification and effects of heavy-tailed noises.

In this article, a modified matrix-weighted fusion estimator is investigated to realize complementary advantages of IMU and UWB for an indoor positioning system in NLOS environments. The main contributions of this article are shown as follows.

1) A NLOS identification method and a binary function are presented to mitigate influence of outliers caused by multipath and NLOS effects for the positioning system.
2) A local modified fusion estimator is designed to fuse the measurements of IMU and UWB by considering both NLOS identification and heavy-tailed noises. A modified matrix-weighted fusion estimator is proposed to improve the estimation accuracy.
3) Sufficient conditions are given to ensure exponentially bounded in mean square of the fusion estimation error.

The remainder of this article is arranged as follows. Section II describes an experimental platform and basic principles of positioning based on IMU and UWB, and puts forward the framework of the positioning system. The local modified fusion estimator and the modified matrix-weighted fusion estimator are designed in Section III. The boundedness of the estimation error is given in Section IV. In Section V, the experimental results are shown, and the positioning performances of different methods are analyzed. This article is concluded in Section VI.

Notation: $\mathbb{R}^{m\times n}$ and $\mathbb{R}^n$ denote $m\times n$ real matrices and sets of n-dimensional real vectors, respectively. I_n represents the n-dimensional identity matrix. M^{T}, M^{-1}, and $\det(M)$ mean the transpose, inverse, and determinant of matrix M, respectively. $E[M]$ and $E[M|G]$ mean the expectation of M and the expectation of M conditional on G. $\|\cdot\|$ denotes the spectral norm of matrices. $\lambda_{\min}(M)$ and $\lambda_{\max}(M)$ represent the minimum and maximum values of the eigenvalues of the symmetric matrix M.

II. Problem Formulation

A. Experimental Platform

In this article, experiments are performed with an experimental platform shown in Fig. 1, where Microprocessor Unit 6050 (MPU_6050) is adopted as the IMU sensor, and the sampling frequency is 50 Hz. The LinkTrack S-V4.0.2 UWB has a communication distance of 80 m, and the rate of updates by it is 50 Hz. In order to simplify the model, position of the target is considered only in the two-dimensional (2-D) plane.

B. Positioning Principle Based on IMU

IMU composed of an accelerometer and a gyroscope is often used to resolve the problem of target positioning in inertial navigation system. Assume that the body coordinate system (the B system) is represented by $O_x_by_bz_b$ and the navigation coordinate system (the N system) is represented by $O_x_ny_nz_n$ as shown in Fig. 2. The N system coincides with the B system

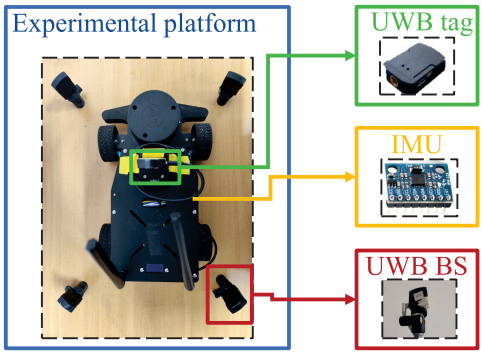


Fig. 1. Experimental platform.

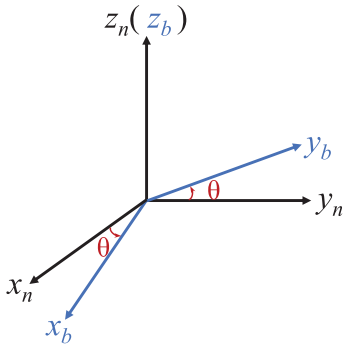


Fig. 2. Rotation transformation of IMU.

through rotating the N system counterclockwise around the z_n-axis at an attitude angle θ.

In the system, the attitude angle θ is usually adopted to calculate the rotation matrix C_b^n from the B system to the N system, where C_b^n is presented as

$$C_b^n = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

The acceleration vector a_b in the B system is expressed as $a_b = [a_x^b a_y^b]^{\mathrm{T}}$, where a_x^b and a_y^b are accelerations measured by the accelerometer in the x_b-axis direction and y_b-axis direction, respectively. Thereby, the acceleration vector a_n in the N system can be obtained as follows:

$$a_n = \begin{bmatrix} a_x^n \\ a_y^n \end{bmatrix} = C_b^n a_b$$

where a_x^n and a_y^n denote accelerations of the target in the x_n-axis direction and y_n-axis direction in the N system, respectively. At this point, a_b measured by the accelerometer is transformed into a_n via the rotation matrix C_b^n. In the following, we will perform the calculation in the N system.

Information measured by IMU is used to calculate the real-time position, velocity and attitude angle of the target by the dead reckoning algorithm. When the sampling time T is short, the target is subjected to approximately constant force in the time interval and does linear motion with uniform acceleration. Let $\vartheta_{x_{k-1}}$ and $\vartheta_{y_{k-1}}$ denote velocities of the target at time $k-1$ in the x_n-axis direction and y_n-axis direction, respectively. $a_{x_{k-1}}^n$ and $a_{y_{k-1}}^n$ are accelerations of the target at time $k-1$ in

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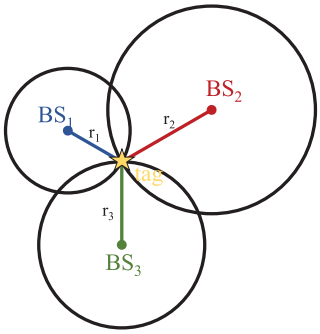


Fig. 3. Multilateral positioning method of UWB.

the x_n-axis direction and y_n-axis direction, respectively. Here, k is the discrete time index. Then, velocities ϑ_{x_k} and ϑ_{y_k} of the target at time k are obtained as

$$\begin{cases} \vartheta_{x_k} = \vartheta_{x_{k-1}} + a^n_{x_{k-1}} T \\ \vartheta_{y_k} = \vartheta_{y_{k-1}} + a^n_{y_{k-1}} T. \end{cases} \tag{1}$$

Further, let $p_{x_{k-1}}$ and $p_{y_{k-1}}$ be positions of the target at time $k-1$ in the x_n-axis direction and y_n-axis direction, respectively. Positions p_{x_k} and p_{y_k} of the target at time k are given as follows:

$$\begin{cases} p_{x_k} = p_{x_{k-1}} + \vartheta_{x_{k-1}} T + \frac{1}{2} a^n_{x_{k-1}} T^2 \\ p_{y_k} = p_{y_{k-1}} + \vartheta_{y_{k-1}} T + \frac{1}{2} a^n_{y_{k-1}} T^2. \end{cases} \tag{2}$$

In addition, denote θ_{k-1} as the attitude angle of the target at time $k-1$, and ω_{k-1} as the angular velocity of the target measured by gyroscope at time $k-1$. Then, the attitude angle θ_k of the target at time k can be described as

$$\theta_k = \theta_{k-1} + \omega_{k-1} T. \tag{3}$$

C. Positioning Principle Based on UWB

Nanosecond or subnanosecond pulses are used to achieve wireless communication in UWB technology, which is a low-power radio technology used in the field of wireless communication. The signal transmission time between the base station (BS) and the target is usually harnessed to calculate the distance, and then the multilateral positioning method is utilized to calculate the target position as shown in Fig. 3.

Through the geometric relationships between BSs and the target in Fig. 3, relationships between BSs and the target are shown in the following equations:

$$\begin{cases} \iota^2_{1,k} = \left(x_{BS_1} - x_{p_k}\right)^2 + \left(y_{BS_1} - y_{p_k}\right)^2 \\ \iota^2_{2,k} = \left(x_{BS_2} - x_{p_k}\right)^2 + \left(y_{BS_2} - y_{p_k}\right)^2 \\ \iota^2_{3,k} = \left(x_{BS_3} - x_{p_k}\right)^2 + \left(y_{BS_3} - y_{p_k}\right)^2 \end{cases} \tag{4}$$

where $\iota_{j,k}, j = 1, 2, 3$ is the measured distance between the jth BS and the target at time k, (x_{BS_j}, y_{BS_j}) is the location of the jth BS, and (x_{p_k}, y_{p_k}) is the location of the target at time k. Further, (x_{BS_j}, y_{BS_j}) is known, and $\iota_{j,k}$ can be measured by UWB. Note that the location of the target (x_{p_k}, y_{p_k}) can be obtained by (4) directly, which will be used in the following system model.

D. System Model Based on IMU and UWB

In practical engineering, it is often not possible to directly obtain the real value of the required state variables due to problems of random interference. In the process of measuring with sensors, observed signals are usually mixed with random noises. Outliers appear too many times for the multipath and NLOS effects, when UWB measures the distance ι_{j_k}. Thus, the process noise and measurement noise have characteristics of heavy-tailed. According to (1)–(3), the state equation is described as follows:

$$\begin{cases} p_{x_k} = p_{x_{k-1}} + \vartheta_{x_{k-1}} T + \frac{1}{2} a^n_{x_{k-1}} T^2 + w_{1_{k-1}} \\ p_{y_k} = p_{y_{k-1}} + \vartheta_{y_{k-1}} T + \frac{1}{2} a^n_{y_{k-1}} T^2 + w_{2_{k-1}} \\ \vartheta_{x_k} = \vartheta_{x_{k-1}} + a^n_{x_{k-1}} T + w_{3_{k-1}} \\ \vartheta_{y_k} = \vartheta_{y_{k-1}} + a^n_{y_{k-1}} T + w_{4_{k-1}} \\ \theta_k = \theta_{k-1} + \omega_{k-1} T + w_{5_{k-1}} \end{cases} \tag{5}$$

where $w_{1_{k-1}}, w_{2_{k-1}}, w_{3_{k-1}}, w_{4_{k-1}}$, and $w_{5_{k-1}}$ denote the process noises. Then, the location of the target (x_{p_k}, y_{p_k}) can be obtained by (4). Three BSs are needed to get a location of the target, and more than three groups of BSs can obtain multiple groups of measured location information. Assume that there are L groups of measured location information, and define the ith group of measured location information at time k as $(x_{p_{i,k}}, y_{p_{i,k}})$. The measurement vector $y_{i,k}$ of the ith group at time k is defined as

$$y_{i,k} = \begin{bmatrix} x_{p_{i,k}} \\ y_{p_{i,k}} \end{bmatrix} = \begin{bmatrix} p_{x_k} + v_{x_{i,k}} \\ p_{y_k} + v_{y_{i,k}} \end{bmatrix}, \ i = 1, 2, \ldots, L \tag{6}$$

where $v_{x_{i,k}}$ and $v_{y_{i,k}}$ are measurement noises.

As a result, define the state vector x_k at time k as $x_k = [p_{x_k} p_{y_k} \vartheta_{x_k} \vartheta_{y_k} \theta_k]^{\mathrm{T}}$. By (5) and (6), the state equation and the measurement equation in matrix form are expressed as follows:

$$\begin{cases} x_k = F x_{k-1} + B u_{k-1} + w_{k-1} \\ y_{i,k} = H_i x_k + v_{i,k}, \ i = 1, 2, \ldots, L \end{cases} \tag{7}$$

where

$$F = \begin{bmatrix} 1 & 0 & T & 0 & 0 \\ 0 & 1 & 0 & T & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \ B = \begin{bmatrix} \frac{T^2}{2} & 0 & 0 \\ 0 & \frac{T^2}{2} & 0 \\ T & 0 & 0 \\ 0 & T & 0 \\ 0 & 0 & T \end{bmatrix}$$

$$H_i = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \ v_{i,k} = \begin{bmatrix} v_{x_{i,k}} \\ v_{y_{i,k}} \end{bmatrix}$$

$$u_k = \left[a_{x_k} a_{y_k} \omega_k\right]^{\mathrm{T}}, \ w_k = \left[w_{1_k} w_{2_k} w_{3_k} w_{4_k} w_{5_k}\right]^{\mathrm{T}}.$$

Besides, it is assumed that the initial state vector x_0, process noise w_k and measurement noise $v_{i,k}$ are uncorrelated and obey non-Gaussian heavy-tailed distribution, which are modeled as the following Student's t distribution:

$$\begin{aligned} p(x_0) &= St\left(x_0; \hat{x}_{0|0}, P_{0|0}, \nu_0\right) \\ p(w_k) &= St(w_k; 0, Q_k, \nu_w) \\ p\left(v_{i,k}\right) &= St\left(v_{i,k}; 0, R_{i,k}, \nu_i\right) \end{aligned}$$

where $St(\cdot; \varrho, \Lambda, \nu)$ is the Student's t distribution whose mean is ϱ, the scale matrix is Λ, and degrees of freedom is ν.

The following lemmas are shown to promote follow-up analysis.

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Lemma 1 [25]: Let $St(x_k;\mu_k,\Theta_k,\nu)$ denote the probability density function of $x_k \in \mathbb{R}^d$ that follows multivariate t-distribution as:

$$p(x_k) = \Xi_k G_1(x_k)$$

where

$$\Xi_k = \frac{\Gamma\left(\frac{\nu+2}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)} \frac{1}{(\nu\pi)^{d/2}} \frac{1}{\sqrt{\det[\Theta_k]}}$$
$$G_1(x_k) = \left(1+\frac{\sigma_k}{\nu}\right)^{-\frac{\nu+2}{2}}$$

with $\sigma_k = [x_k-\mu_k]^{\mathrm{T}}\Theta_k^{-1}[x_k-\mu_k]$. Probability density function which follows multivariate t-distribution has the following properties: 1) if $z_k = Ax_k+B$, then probability density function of z_k is $p(z_k) = St(z_k; A\mu_k+B, A\Theta_k A^{\mathrm{T}}, \nu)$ and 2) if random vectors $z_{1,k} \in \mathbb{R}^{d_1}$ and $z_{2,k} \in \mathbb{R}^{d_2}$ obey the multivariate t-distribution.

Then, the probability density function of them are shown as

$$p(z_{1,k}) = St(z_{1,k};\mu_{1,k},\Theta_{11,k},\nu)$$
$$p(z_{2,k}) = St(z_{2,k};\mu_{2,k},\Theta_{22,k},\nu).$$

Then, the joint t-distribution of $z_{1,k}$ and $z_{2,k}$ is written as

$$p(z_{1,k},z_{2,k}) = St\left(\begin{bmatrix} z_{1,k} \\ z_{2,k}\end{bmatrix}; \begin{bmatrix}\mu_{1,k}\\ \mu_{2,k}\end{bmatrix}, \begin{bmatrix}\Theta_{11,k} & \Theta_{12,k}\\ \Theta_{21,k} & \Theta_{22,k}\end{bmatrix}, \nu\right)$$

where $\Theta_{12,k} \in \mathbb{R}^{d_1\times d_2}$ and $\Theta_{21,k} \in \mathbb{R}^{d_2\times d_1}$ are joint scale matrices of $z_{1,k}$ and $z_{2,k}$.

3) The conditional probability density function is shown as

$$p(z_{1,k}|z_{2,k}) = St(z_{1,k};\mu_{1|2,k},\Theta_{1|2,k},\nu_{1|2})$$

where

$$\nu_{1|2} = \nu + d_2$$
$$\mu_{1|2,k} = \mu_{1,k} + \Theta_{12,k}\Theta_{22,k}^{-1}[z_{2,k}-\mu_{2,k}]$$
$$\Theta_{1|2,k} = \frac{\nu+\sigma_{2,k}}{\nu+d_2}\left[\Theta_{11,k}-\Theta_{12,k}\Theta_{22,k}^{-1}\Theta_{12,k}^{\mathrm{T}}\right]$$

with $\sigma_{2,k} = [z_{2,k}-\mu_{2,k}]^{\mathrm{T}}\Theta_{22,k}^{-1}[z_{2,k}-\mu_{2,k}]$.

Lemma 2 [27]: For random vectors $p, q \in \mathbb{R}^n$, the following inequality satisfies:

$$pq^{\mathrm{T}} + qp^{\mathrm{T}} \le \varpi pp^{\mathrm{T}} + \varpi^{-1}qq^{\mathrm{T}}$$

where ϖ is a scalar.

Lemma 3 [28]: If there exists a symmetric matrix $\Lambda \in \mathbb{R}^{n\times n}$, then the following inequality exists:

$$\lambda_{\min}(\Lambda)p^{\mathrm{T}}p \le p^{\mathrm{T}}\Lambda p \le \lambda_{\max}(\Lambda)p^{\mathrm{T}}p.$$

Lemma 4 [29]: Considering a discrete stochastic process z_t, assume that there exists a random function $V_t(z_t)$ and scalars $0 < t_1 \le 1$, $t_2, t_3 > 0$ satisfying $t_3\|z_t\|^2 \le V_t(z_t)$ and $E[V_{t+1}(z_{t+1})|z_t] - V_t(z_t) \le -t_1V_t(z_t)+t_2$. Then the Lyapunov function $V_t(z_t)$ and the stochastic process z_t are mean square exponentially bounded.

Remark 1: Note that the object target is a wheeled robot which moves only in the 2-D plane. Therefore, only the case of rotation around the z_n-axis is considered when describing positioning principle of the IMU.

III. Main Result

In this section, a local modified fusion estimator based on IMU and UWB is proposed. Through KF framework, the ith group of location information $y_{i,k}$ of UWB at time k is fused with the information measured by IMU to obtain the local estimation $\hat{x}_{i,k|k}$ and the scale matrix of the estimation error $P_{i,k|k}$. Furthermore, inspired by the matrix-weighted fusion algorithm, a modified matrix-weighted fusion estimator is designed. After obtaining L groups of the local estimation $\hat{x}_{i,k|k}$ and the scale matrix of the estimation error $P_{i,k|k}$, a modified matrix-weighted fusion estimation $\hat{x}_{F,k|k}$ is gotten.

A. Local Modified Fusion Estimator

In this part, a local modified fusion estimator is designed in NLOS environments, which fuses the measurements of IMU and UWB to locate the target. In addition, a NLOS identification method is adopted as follows:

$$\begin{aligned}\Delta r_{i,k} &= |r_{i,k} - r_{i,k-\mu}| \\ &= \begin{cases} |d_{i,k} - d_{i,k-\mu} + \varepsilon_{i,\mathrm{LOS}}|, & \mathrm{LOS}_{k-\mu}\to\mathrm{LOS}_k \\ |d_{i,k} - d_{i,k-\mu} + \varepsilon_{i,\mathrm{NLOS}}|, & \mathrm{LOS}_{k-\mu}\to\mathrm{NLOS}_k \end{cases}\end{aligned} \tag{8}$$

where $r_{i,k} = \sqrt{(x_{p_{i,k}})^2 + (y_{p_{i,k}})^2}$, $\Delta r_{i,k}$ is the absolute value of the difference between $r_{i,k}$ and $r_{i,k-\mu}$, and $d_{i,k}$ is the true distance between the ith BS and the target at time k. $\mathrm{LOS}_{k-\mu}\to\mathrm{NLOS}_k$ represents the measurement environment from the line-of-sight (LOS) situation at time $k-\mu$ to the NLOS situation at time k. Considering that NLOS effects do not disappear in one period, that is, the outliers appear at consecutive multiple moments. Therefore, different from existing results, μ is a constant greater than one in the NLOS identification method designed in this article.

Let $\Delta d_{i,k} = |d_{i,k} - d_{i,k-\mu}|$. In fact, there is a deviation between $\Delta r_{i,k}$ and $\Delta d_{i,k}$ due to external factors. In the NLOS situation, the deviation $\varepsilon_{i,\mathrm{NLOS}}$ is large due to the signal being occluded. In the LOS situation, the deviation $\varepsilon_{i,\mathrm{LOS}}$ is relatively small. Furthermore, in this article, $\Delta d_{i,k}$ is a small value due to the speed limitation, and the sampling period of 50 HZ. Thereby, if it is the LOS situation, $\Delta r_{i,k}$ is a relatively small value. If it is the NLOS situation, then the deviation $\varepsilon_{i,\mathrm{NLOS}}$ is going to be larger, and correspondingly $\Delta r_{i,k}$ becomes larger. As a result, we can use $\Delta r_{i,k}$ to identify whether the target enters the NLOS environments.

In order to indicate whether the target is in NLOS environments, for the ith group of location information, the following binary function is given by:

$$\gamma_{i,k} = \begin{cases} 0, \text{ if } \Delta r_{i,k} > \alpha_i \\ 1, \text{ otherwise}\end{cases} \tag{9}$$

where α_i is a constant, which is adjustable. Besides, $\gamma_{i,k} = 0$ represents the situation of $\mathrm{LOS}_{k-\mu}\to\mathrm{NLOS}_k$ in (8). When $\gamma_{i,k} = 1$, the current information measured by UWB is used by the local modified fusion estimator. When $\gamma_{i,k} = 0$, the information measured by UWB is not used by the local modified fusion estimator. Thereby, similar to [25], the sequence of measurement information received by the local fusion estimator is expressed as follows:

$$Y_{i,k} = \{\gamma_{i,0}y_{i,0}, \ldots, \gamma_{i,k}y_{i,k}\}.$$

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As shown in the following theorem, the designed local modified fusion estimator is given. Note that, in this article, the sampling time of UWB is consistent with IMU.

Theorem 1: Consider the model (7) as well as the NLOS identification method (9). Under the following condition as:

$$p\left(x_{i,k}|Y_{i,k-1}\right) = St\left(x_{i,k}; \hat{x}_{i,k|k-1}, P_{i,k|k-1}, \nu_0\right)$$

the local modified fusion estimator is given as follows.

1) *Time Update:*

$$\begin{aligned}\hat{x}_{i,k|k-1} &= F\hat{x}_{i,k-1|k-1} + Bu_{k-1}\\ P_{i,k|k-1} &= FP_{i,k-1|k-1}F^{\mathrm{T}} + \frac{\nu_w(\nu_0-2)}{(\nu_w-2)\nu_0}Q_{k-1}.\end{aligned}$$

2) *Measurement Information Update:*

$$\hat{x}_{i,k|k} = \hat{x}_{i,k|k-1} + \gamma_{i,k}K_{i,k}\tilde{y}_{i,k|k-1} \tag{10}$$

$$\begin{aligned}P_{i,k|k} = {}& \gamma_{i,k}\Omega_{i,k}\left(P_{i,k|k-1} - K_{i,k}P^{\tilde{y}\tilde{y}}_{i,k|k-1}K^{\mathrm{T}}_{i,k}\right)\\ &+ \left(1-\gamma_{i,k}\right)P_{i,k|k-1}\end{aligned} \tag{11}$$

where

$$\begin{aligned}K_{i,k} &= P_{i,k|k-1}H_i^{\mathrm{T}}\left(P^{\tilde{y}\tilde{y}}_{i,k|k-1}\right)^{-1}\\ P^{\tilde{y}\tilde{y}}_{i,k|k-1} &= H_iP_{i,k|k-1}H_i^{\mathrm{T}} + \frac{\nu_i(\nu_0-2)}{(\nu_i-2)\nu_0}R_{i,k}\\ \tilde{y}_{i,k|k-1} &= y_{i,k} - H_i\hat{x}_{i,k|k-1}\\ \Omega_{i,k} &= \frac{(\nu_0-2)\left(\nu_0 + o^2_{i,k}\right)}{\nu_0(\nu_0+m-2)}\end{aligned}$$

with $o^2_{i,k} = \tilde{y}^{\mathrm{T}}_{i,k|k-1}(P^{\tilde{y}\tilde{y}}_{i,k|k-1})^{-1}\tilde{y}_{i,k|k-1}$.

Proof: Inspired by [24], the state prediction at state update time k is computed as follows:

$$\begin{aligned}\hat{x}_{i,k|k-1} &= E\left[x_{i,k}|Y_{i,k-1}\right]\\ &= F\hat{x}_{i,k-1|k-1} + Bu_{k-1}.\end{aligned} \tag{12}$$

Define $\tilde{x}_{i,k|k-1} = x_k - \hat{x}_{i,k|k-1}$ as the state prediction error vector. Then, the scale matrix of the prediction error can be obtained as

$$\begin{aligned}P_{i,k|k-1} &= \frac{\nu_0-2}{\nu_0}E\left[\tilde{x}_{i,k|k-1}\tilde{x}^{\mathrm{T}}_{i,k|k-1}|Y_{i,k-1}\right]\\ &= FP_{i,k-1|k-1}F^{\mathrm{T}} + \frac{\nu_w(\nu_0-2)}{(\nu_w-2)\nu_0}Q_{k-1}.\end{aligned}$$

By (12), we have

$$\begin{aligned}\hat{y}_{i,k|k-1} &= E\left[y_{i,k}|Y_{i,k-1}\right]\\ &= H_i\hat{x}_{i,k|k-1}.\end{aligned}$$

Then, the scale matrix of the measurement prediction error is gotten by

$$\begin{aligned}P^{\tilde{y}\tilde{y}}_{i,k|k-1} &= E\left[\tilde{y}_{i,k|k-1}\tilde{y}^{\mathrm{T}}_{i,k|k-1}|Y_{i,k-1}\right]\\ &= H_iP_{i,k|k-1}H_i^{\mathrm{T}} + \frac{\nu_i(\nu_0-2)}{(\nu_i-2)\nu_0}R_{i,k}.\end{aligned}$$

Based on Lemma 1, there exists

$$\begin{aligned}P^{\tilde{x}\tilde{y}}_{i,k|k-1} &= E\left[\tilde{x}_{i,k|k-1}\tilde{y}^{\mathrm{T}}_{i,k|k-1}|Y_{i,k-1}\right]\\ &= P_{i,k|k-1}H_i^{\mathrm{T}}.\end{aligned}$$

Besides, the Kalman gain matrix $K_{i,k}$ is shown as

$$K_{i,k} = P^{\tilde{x}\tilde{y}}_{i,k|k-1}\left(P^{\tilde{y}\tilde{y}}_{i,k|k-1}\right)^{-1}. \tag{13}$$

In the measurement update, there are two cases according to the NLOS identification method shown as (9).

Case 1 ($\gamma_{i,k} = 1$): The ith local fusion estimator receives measurement $y_{i,k}$ at time k, that is

$$Y_{i,k} = \{\gamma_{i,0}y_{i,0}, \ldots, \gamma_{i,k-1}y_{i,k-1}, y_{i,k}\}.$$

According to Lemma 1, the conditional probability density function is given as

$$p\left(x_{i,k}|Y_{i,k}\right) = \frac{p\left(x_{i,k}, y_{i,k}|Y_{i,k-1}\right)}{p\left(y_{i,k}|Y_{i,k-1}\right)} = St\left(x_{i,k}; \hat{x}'_{i,k|k}, P'_{i,k|k}, \nu'_0\right)$$

where the jointly predicted probability density function $p(x_{i,k}, y_{i,k}|Y_{i,k-1})$ and the likelihood predicted probability density function $p(y_{i,k}|Y_{i,k-1})$ obey the Student's t distribution.

In light of Lemma 1 and (13), the local state estimation is expressed as

$$\begin{aligned}\hat{x}_{i,k|k} &= \hat{x}'_{i,k|k}\\ &= E\left[x_{i,k}|Y_{i,k}\right]\\ &= \hat{x}_{i,k|k-1} + P^{\tilde{x}\tilde{y}}_{i,k|k-1}\left(P^{\tilde{y}\tilde{y}}_{i,k|k-1}\right)^{-1}\tilde{y}_{i,k|k-1}\\ &= \hat{x}_{i,k|k-1} + K_{i,k}\tilde{y}_{i,k|k-1}.\end{aligned} \tag{14}$$

Define $\bar{x}_{i,k|k} = x_k - \hat{x}_{i,k|k}$ as the state error vector. Then, the scale matrix of the estimation error can be obtained as follows:

$$\begin{aligned}P_{i,k|k} &= \frac{\nu'_0(\nu_0-2)}{\left(\nu'_0-2\right)\nu_0}P'_{i,k|k}\\ &= \frac{\nu_0-2}{\nu_0}E\left[\bar{x}_{i,k|k}\bar{x}^{\mathrm{T}}_{i,k|k}|Y_{i,k}\right]\\ &= \Omega_{i,k}\left(P_{i,k|k-1} - K_{i,k}P^{\tilde{y}\tilde{y}}_{i,k|k-1}K^{\mathrm{T}}_{i,k}\right)\end{aligned} \tag{15}$$

where $\nu'_0 = \nu_0 + m$. Moreover, m denotes the dimension of the measurement vector $y_{i,k}$.

Case 2 ($\gamma_{i,k} = 0$): The ith local fusion estimator does not receive the measurement $y_{i,k}$ at time k. The measurement update is no longer performed in this case.

Meanwhile, the state prediction value is used as the state estimation, and the scale matrix of the prediction error is used as the scale matrix of the estimation error. That is, the local state estimation is

$$\hat{x}_{i,k|k} = \hat{x}_{i,k|k-1}. \tag{16}$$

The scale matrix of the estimation error is expressed as

$$P_{i,k|k} = P_{i,k|k-1}. \tag{17}$$

Therefore, the local state estimation in (10) is computed from (14) and (16), and the scale matrix of the estimation error in (11) is obtained from (15) and (17). The proof is completed. ■

Remark 2: The choice of NLOS identification method parameter α_i should consider both drift of IMU and the outliers caused by NLOS. With decrease of α_i, NLOS identification

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effect is obvious and the state estimation becomes uncertain and incredible due to the random drift of IMU error accumulation, the variety of tasks, and other negative factors. Therefore, it is important to choice an appropriate parameter α_i for better state estimation.

B. Modified Matrix-Weighted Fusion Estimator

In this part, a modified matrix-weighted fusion estimator is obtained for high-accuracy estimation and robustness. Local state prediction error vector $\tilde{x}_{i,k|k-1}$ and local state error vector $\bar{x}_{i,k|k}$ are expressed as follows:

$$\begin{aligned}\tilde{x}_{i,k|k-1} &= F\bar{x}_{i,k-1|k-1} + w_{k-1} \\ \bar{x}_{i,k|k} &= \left(I_n - \gamma_{i,k}K_{i,k}H_i\right)\tilde{x}_{i,k|k-1} - \gamma_{i,k}K_{i,k}v_{i,k}\end{aligned} \tag{18}$$

where n denotes the dimension of the state vector x_k. Inspired by [30], the prediction error scale matrix between the ith and the jth sensor subsystems is derived as

$$\begin{aligned}P_{ij,k|k-1} &= E\left[\tilde{x}_{i,k|k-1}\tilde{x}_{j,k|k-1}^{\mathrm{T}}\right] \\ &= FP_{ij,k-1|k-1}F^{\mathrm{T}} + Q_{k-1}.\end{aligned}$$

The error scale matrix between the ith and the jth sensor subsystems is given by

$$\begin{aligned}P_{ij,k|k} &= E\left[\bar{x}_{i,k|k}\bar{x}_{j,k|k}^{\mathrm{T}}\right] \\ &= \left(I_n - \gamma_{i,k}K_{i,k}H_i\right)P_{ij,k|k-1}\left(I_n - \gamma_{j,k}K_{j,k}H_j\right)^{\mathrm{T}}.\end{aligned}$$

When there are L local modified fusion estimators, the modified matrix-weighted fusion estimation $\hat{x}_{F,k|k}$ and its covariance $P_{F,k|k}$ are shown as

$$\begin{aligned}\hat{x}_{F,k|k} &= \sum_{i=1}^{L} A_{i,k}\hat{x}_{i,k|k} \\ &= A_{1,k}\hat{x}_{1,k|k} + A_{2,k}\hat{x}_{2,k|k} + \cdots + A_{L,k}\hat{x}_{L,k|k} \\ P_{F,k|k} &= \left(e^{\mathrm{T}}\Sigma_k^{-1}e\right)^{-1}\end{aligned}$$

where

$$\begin{aligned}A_{F,k} &= \left[A_{1,k}, A_{2,k}, \ldots, A_{L,k}\right]^{\mathrm{T}} \\ &= \Sigma_k^{-1}e\left(e^{\mathrm{T}}\Sigma_k^{-1}e\right)^{-1} \\ e &= \left[I_n, \ldots, I_n\right]^{\mathrm{T}}\end{aligned}$$

$$\Sigma_k = \begin{bmatrix} P_{11,k|k} & P_{12,k|k} & \cdots & P_{1L,k|k} \\ P_{21,k|k} & P_{22,k|k} & \cdots & P_{2L,k|k} \\ \vdots & \vdots & \ddots & \vdots \\ P_{L1,k|k} & P_{L2,k|k} & \cdots & P_{LL,k|k} \end{bmatrix}.$$

Furthermore, $\Sigma_k = (P_{ij})_{nL\times nL}$, $i, j = 1, 2, \ldots, L$, is a $nL \times nL$ positive definite matrix, and $A_{F,k}$ and e are both $nL \times n$ matrices.

Remark 3: Note that a matrix-weighted fusion estimator is presented to obtain the global state, which is robust and flexible due to parallel processing framework [31]. Moreover, the weight of each sensor can be adjusted more carefully in the fusion process.

IV. Stability Analysis

In this part, the following theorem is given for the fusion estimation error of the modified matrix-weighted fusion estimator.

Theorem 2: For the model (7), if there exist positive numbers $\kappa_1, \kappa_2, \kappa_3, \kappa_4, \sigma, \varsigma$, and a matrix $\Pi_i > 0$ satisfying the following inequalities:

$$\begin{aligned}\sigma I_n > {} & \gamma_{i,k+1}^2\left(1 + \kappa_1^{-1} + \kappa_3\right)H_i^{\mathrm{T}}K_{i,k+1}^{\mathrm{T}}A_{i,k+1}^{\mathrm{T}}\Pi_i A_{i,k+1} \\ & \times K_{i,k+1}H_i \end{aligned} \tag{19}$$

$$\begin{aligned}-\varsigma I_n > {} & \left[(1+\kappa_1+\kappa_2)(1+\kappa_4)F^{\mathrm{T}}F - I_n\right]A_{i,k+1}^{\mathrm{T}}\Pi_i \\ & \times A_{i,k+1} + \sigma(1+\kappa_4)F^{\mathrm{T}}F \end{aligned} \tag{20}$$

$$1 > \frac{\varsigma}{\lambda_{\max}\left(A_{i,k+1}^{\mathrm{T}}\Pi_i A_{i,k+1}\right)} > 0 \tag{21}$$

then the fusion estimation error is mean square exponentially bounded.

Proof: The fusion estimation error $\bar{x}_{F,k}$ is shown as

$$\begin{aligned}\bar{x}_{F,k} &= x_k - \hat{x}_{F,k} \\ &= \sum_{i=1}^{L} A_{i,k}\bar{x}_{i,k|k} \\ &= A_{1,k}\bar{x}_{1,k|k} + A_{2,k}\bar{x}_{2,k|k} + \cdots + A_{L,k}\bar{x}_{L,k|k}.\end{aligned} \tag{22}$$

Next, select the Lyapunov function as follows:

$$\begin{aligned}V_k\left(\bar{x}_{F,k}\right) &= \bar{x}_{F,k}^{\mathrm{T}}\Pi\bar{x}_{F,k} \\ &= \sum_{i=1}^{L}\bar{x}_{i,k|k}^{\mathrm{T}}A_{i,k}^{\mathrm{T}}\Pi_i A_{i,k}\bar{x}_{i,k|k}\end{aligned} \tag{23}$$

where Π_i is a positive definite symmetric matrix, the block diagonal matrix $\Pi = \mathrm{diag}\{\Pi_1, \ldots, \Pi_L\}$. Then, we have

$$\Delta V_k\left(\bar{x}_{F,k}\right) = E\left[V_{k+1}\left(\bar{x}_{F,k+1}\right)|\bar{x}_{F,k}\right] - V_k\left(\bar{x}_{F,k}\right). \tag{24}$$

Substituting (22) and (23) into (24), it is shown as

$$\begin{aligned}\Delta V_k\left(\bar{x}_{F,k}\right) = \sum_{i=1}^{L} & [\tilde{x}_{i,k+1|k}^{\mathrm{T}}\Gamma_{1i,k+1}\tilde{x}_{i,k+1|k} \\ & - \tilde{x}_{i,k+1|k}^{\mathrm{T}}\Gamma_{1i,k+1}\gamma_{i,k+1}K_{i,k+1}H_i\tilde{x}_{i,k+1|k} \\ & - \tilde{x}_{i,k+1|k}^{\mathrm{T}}\Gamma_{1i,k+1}\gamma_{i,k+1}K_{i,k+1}v_{i,k+1} \\ & - \tilde{x}_{i,k+1|k}^{\mathrm{T}}H_i^{\mathrm{T}}K_{i,k+1}^{\mathrm{T}}\gamma_{i,k+1}^{\mathrm{T}}\Gamma_{1i,k+1}\tilde{x}_{i,k+1|k} \\ & + \tilde{x}_{i,k+1|k}^{\mathrm{T}}H_i^{\mathrm{T}}K_{i,k+1}^{\mathrm{T}}\gamma_{i,k+1}^{\mathrm{T}}\Gamma_{1i,k+1}\gamma_{i,k+1}K_{i,k+1}H_i \\ & \times \tilde{x}_{i,k+1|k} \\ & + \tilde{x}_{i,k+1|k}^{\mathrm{T}}H_i^{\mathrm{T}}K_{i,k+1}^{\mathrm{T}}\gamma_{i,k+1}^{\mathrm{T}}\Gamma_{1i,k+1}\gamma_{i,k+1}K_{i,k+1}v_{i,k+1} \\ & - v_{i,k+1}^{\mathrm{T}}K_{i,k+1}^{\mathrm{T}}\gamma_{1i,k+1}^{\mathrm{T}}\Gamma_{1i,k+1}\tilde{x}_{i,k+1|k} \\ & + v_{i,k+1}^{\mathrm{T}}K_{i,k+1}^{\mathrm{T}}\gamma_{1i,k+1}^{\mathrm{T}}\Gamma_{1i,k+1}\gamma_{i,k+1}K_{i,k+1}H_i\tilde{x}_{i,k+1|k} \\ & + v_{i,k+1}^{\mathrm{T}}K_{i,k+1}^{\mathrm{T}}\gamma_{1i,k+1}^{\mathrm{T}}\Gamma_{1i,k+1}\gamma_{i,k+1}K_{i,k+1}v_{i,k+1} \\ & - \bar{x}_{i,k|k}^{\mathrm{T}}A_{i,k}^{\mathrm{T}}\Pi_i A_{i,k}\bar{x}_{i,k|k}]\end{aligned}$$

where $\Gamma_{1i,k+1} = A_{i,k+1}^{\mathrm{T}}\Pi_i A_{i,k+1}$.

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Based on Lemmas 2 and 3, we have

$$\begin{aligned}\Delta V_k(\bar{x}_{F,k}) \le \sum_{i=1}^{L}\big[&(1+\kappa_1+\kappa_2)\lambda_{\max}(\Gamma_{1i,k+1})\|\tilde{x}_{i,k+1|k}\|^2\\&+\gamma_{i,k+1}^2\Big(1+\kappa_1^{-1}+\kappa_3\Big)\lambda_{\max}(\Gamma_{2i,k+1})\|\tilde{x}_{i,k+1|k}\|^2\\&+\gamma_{i,k+1}^2\Big(1+\kappa_2^{-1}+\kappa_3^{-1}\Big)\lambda_{\max}(\Gamma_{3i,k+1})\|v_{i,k+1}\|^2\\&-\bar{x}_{i,k|k}^{\mathrm{T}}A_{i,k}^{\mathrm{T}}\Pi_iA_{i,k}\bar{x}_{i,k|k}\big]\end{aligned}\tag{25}$$

where

$$\begin{aligned}\Gamma_{2i,k+1}&=H_i^{\mathrm{T}}K_{i,k+1}^{\mathrm{T}}\Gamma_{1i,k+1}K_{i,k+1}H_i\\\Gamma_{3i,k+1}&=K_{i,k+1}^{\mathrm{T}}\Gamma_{1i,k+1}K_{i,k+1}.\end{aligned}$$

By the same way, based on (19), one has that

$$\begin{aligned}\|\tilde{x}_{i,k+1|k}\|^2&=\tilde{x}_{i,k+1|k}^{\mathrm{T}}\tilde{x}_{i,k+1|k}\\&=\bar{x}_{i,k|k}^{\mathrm{T}}F^{\mathrm{T}}F\bar{x}_{i,k|k}+w_k^{\mathrm{T}}w_k+\bar{x}_{i,k|k}^{\mathrm{T}}F^{\mathrm{T}}w_k\\&\quad+w_k^{\mathrm{T}}F\bar{x}_{i,k|k}\\&\le(1+\kappa_4)\lambda_{\max}\big(F^{\mathrm{T}}F\big)\|\bar{x}_{i,k|k}\|^2\\&\quad+\Big(1+\kappa_4^{-1}\Big)\|w_k\|^2.\end{aligned}\tag{26}$$

Substituting (26) into (25), it is expressed that

$$\begin{aligned}\Delta V_k(\bar{x}_{F,k}) \le \sum_{i=1}^{L}\Big\{&[(1+\kappa_1+\kappa_2)\lambda_{\max}(\Gamma_{1i,k+1})\\&+\gamma_{i,k+1}^2\Big(1+\kappa_1^{-1}+\kappa_3\Big)\lambda_{\max}(\Gamma_{2i,k+1})]\\&\times[(1+\kappa_4)\lambda_{\max}\big(F^{\mathrm{T}}F\big)\|\bar{x}_{i,k|k}\|^2+\Big(1+\kappa_4^{-1}\Big)\\&\times\|w_k\|^2]+\gamma_{i,k+1}^2\Big(1+\kappa_2^{-1}+\kappa_3^{-1}\Big)\\&\times\lambda_{\max}(\Gamma_{3i,k+1})\|v_{i,k+1}\|^2-\bar{x}_{i,k|k}^{\mathrm{T}}A_{i,k}^{\mathrm{T}}\Pi_iA_{i,k}\bar{x}_{i,k|k}\Big\}.\end{aligned}$$

If conditions (19), (20), and (21) are satisfied, then there exists

$$\Delta V_k(\bar{x}_{F,k}) \le -\sum_{i=1}^{L}\varsigma\|\bar{x}_{i,k|k}\|^2+k_2 \le -k_1V_k(\bar{x}_{F,k})+k_2$$

where

$$\begin{aligned}k_1&=\frac{\varsigma}{\lambda_{\max}\Big(A_{i,k+1}^{\mathrm{T}}\Pi_iA_{i,k+1}\Big)}\\k_2&=\sum_{i=1}^{L}\Big\{\Big[(1+\kappa_1+\kappa_2)\lambda_{\max}\big(A_{i,k+1}^{\mathrm{T}}\Pi_iA_{i,k+1}\big)\\&\quad+\gamma_{i,k+1}^2\Big(1+\kappa_1^{-1}+\kappa_3\Big)\lambda_{\max}(H_i^{\mathrm{T}}K_{i,k+1}^{\mathrm{T}}A_{i,k+1}^{\mathrm{T}}\Pi_i\\&\quad\times A_{i,k+1}K_{i,k+1}H_i)\Big]\\&\quad\times\Big[\Big(1+\kappa_4^{-1}\Big)\lambda_{\max}\|w_k\|^2\Big]\\&\quad+\gamma_{i,k+1}^2\Big(1+\kappa_2^{-1}+\kappa_3^{-1}\Big)\lambda_{\max}(K_{i,k+1}^{\mathrm{T}}A_{i,k+1}^{\mathrm{T}}\Pi_i\\&\quad\times A_{i,k+1}K_{i,k+1})\|v_{i,k+1}\|^2\Big\}.\end{aligned}$$

By using Lemma 4, it is obtained that the fusion estimation error is mean square exponentially bounded. This completes the proof. ■

V. Experiment Results

In this section, two experiments are conducted to testify the performance of Algorithm 1.

A. Experimental Results With Trajectory as Line

In this experiment, a moving trajectory of the target with constant velocity is a straight line. For the model (7), the initial state, which follows the Student's t distribution, is represented as:

$$p(x_0)=St\big(x_0;\hat{x}_{0|0},P_{0|0},\nu_0\big)\tag{27}$$

where $\nu_0=3$, $\hat{x}_{0|0}=[0.90.690.120.12(\pi/4)]^{\mathrm{T}}$, $P_{0|0}=\mathrm{diag}[1,1,1,1,1]$. Note that, $(0.9, 0.69)$ is the location of the target obtained by UWB at time $k=0$. Similarly, the process noise w_k and measurement noise $v_{i,k}$ are satisfied with

$$p(w_k)=St(w_k;0,Q_k,\nu_w)\tag{28}$$

$$p\big(v_{i,k}\big)=St\big(v_{i,k};0,R_{i,k},\nu_i\big)\tag{29}$$

where $Q_k=\mathrm{diag}[0.1,0.1,5,5,0.5]$, $R_{1,k}=\mathrm{diag}[1.35,1.8]$, and $R_{2,k}=\mathrm{diag}[1.35,1.8]$. $\nu_w=5$, $\nu_i=3$, and $P_{F,0|0}=\mathrm{diag}[1,1,1,1,1]$. In addition, the sampling period is set as $T=0.02$ s. The parameters of the NLOS identification method in (8) and (9) are set as $\mu=20$, $\alpha_i=0.21$.

On the one hand, in order to verify the effectiveness of the local modified fusion estimator in NLOS environments, the experimental results are first presented by comparing with that of the standard KF. In addition, another comparison is done between the proposed local fusion and KF considering NLOS identification to reflect the contribution of this article.

Figs. 4 and 5 show the error performance comparisons in the x_n-axis direction and y_n-axis direction, respectively. It is seen that the trajectory accuracy can be significantly improved in NLOS environments by utilizing the proposed local modified fusion algorithm. In Fig. 5, when $k\in(380,415)$, it is interfered by obstacles. The error in x_n-axis direction is over 0.2 m and within 0.15 m by using the KF algorithm and the algorithm of KF considering NLOS identification, respectively. While the error is within 0.1 m by using the proposed local modified fusion algorithm. It is concluded that the proposed local modified fusion algorithm has less uncertainty compared with that of KF considering NLOS identification and KF.

On the other hand, Figs. 6 and 7 show the position performance comparisons of the proposed modified matrix-weighted fusion algorithm and the integration algorithm in [16] in the x_n-axis direction and y_n-axis direction, respectively. It is shown that the measurement outliers appear due to the NLOS effects. From Figs. 6 and 7, it is presented that, compared with the integration algorithm in [16], trajectories estimated by the proposed modified matrix-weighted fusion algorithm are smoother and closer to the true trajectory.

In order to further show the effectiveness of the proposed algorithm, the root mean square error (RMSE) is defined as follows:

$$\mathrm{RMSE}=\sqrt{\frac{\sum_{k=1}^{M}\Big\{\Big[\xi(1,k)-\hat{\xi}(1,k)\Big]^2+\Big[\xi(2,k)-\hat{\xi}(2,k)\Big]^2\Big\}}{M}}$$

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Algorithm 1 Modified Matrix-Weighted Fusion Estimator

Initialize

$$\hat{x}_{F,0|0} = \hat{x}_{0|0} = E[x_0]$$
$$P_{F,0|0} = P_{0|0} = E\Big[\big(x_0 - \hat{x}_{0|0}\big)\big(x_0 - \hat{x}_{0|0}\big)^{\mathrm{T}}\Big]$$

Step 1 NLOS identification.

$$\Delta r_{i,k} = |r_{i,k} - r_{i,k-\mu}|$$
$$\gamma_{i,k} = \begin{cases} 0, \text{ if } \Delta r_{i,k} > \alpha_i \\ 1, \text{ otherwise} \end{cases}$$

where α_i is given scalar.

Step 2 Local modified fusion estimator.

1) *Local Prediction:*

$$\hat{x}_{i,k|k-1} = F\hat{x}_{i,k-1|k-1} + B\mathrm{u}_{k-1}$$
$$P_{i,k|k-1} = FP_{i,k-1|k-1}F^{\mathrm{T}} + \frac{\nu_w(\nu_0-2)}{(\nu_w-2)\nu_0}Q_{k-1}$$

2) *Local Update:*

$$P^{\tilde{y}\tilde{y}}_{i,k|k-1} = H_iP_{i,k|k-1}H_i^{\mathrm{T}} + \frac{\nu_i(\nu_0-2)}{(\nu_i-2)\nu_0}R_{i,k}$$
$$P^{\tilde{x}\tilde{y}}_{i,k|k-1} = P_{i,k|k-1}H_i^{\mathrm{T}}$$
$$K_{i,k} = P^{\tilde{x}\tilde{y}}_{i,k|k-1}\Big(P^{\tilde{y}\tilde{y}}_{i,k|k-1}\Big)^{-1}$$
$$\hat{x}_{i,k|k} = \hat{x}_{i,k|k-1} + \gamma_{i,k}K_{i,k}\tilde{y}_{i,k|k-1}$$
$$P_{i,k|k} = \gamma_{i,k}\Omega_{i,k}\Big(P_{i,k|k-1} - K_{i,k}P^{\tilde{y}\tilde{y}}_{i,k|k-1}K_{i,k}^{\mathrm{T}}\Big) + \big(1-\gamma_{i,k}\big)P_{i,k|k-1}$$

where $\tilde{y}_{i,k|k-1} = y_{i,k} - \hat{y}_{i,k|k-1}$, $\Omega_{i,k} = \frac{(\nu_0-2)(\nu_0+o^2_{i,k})}{\nu_0(\nu_0+m-2)}$, with $o^2_{i,k} = \tilde{y}^{\mathrm{T}}_{i,k|k-1}(P^{\tilde{y}\tilde{y}}_{i,k|k-1})^{-1}\tilde{y}_{i,k|k-1}$, and ν_w, ν_0, ν_i are given scalars.

Step 3 Modified matrix-weighted fusion estimator.

$$\hat{x}_{F,k|k} = \sum_{i=1}^{L} A_{i,k}\hat{x}_{i,k|k}$$
$$P_{F,k|k} = \Big(e^{\mathrm{T}}\Sigma_k^{-1}e\Big)^{-1}$$

where $e = [I_n, \ldots, I_n]^{\mathrm{T}}$, $\Sigma_k = \big(P_{ij}\big)_{nL\times nL}$, and $A_{F,k} = [A_{1,k}, A_{2,k}, \ldots, A_{L,k}]^{\mathrm{T}} = \Sigma_k^{-1}e(e^{\mathrm{T}}\Sigma_k^{-1}e)^{-1}$.

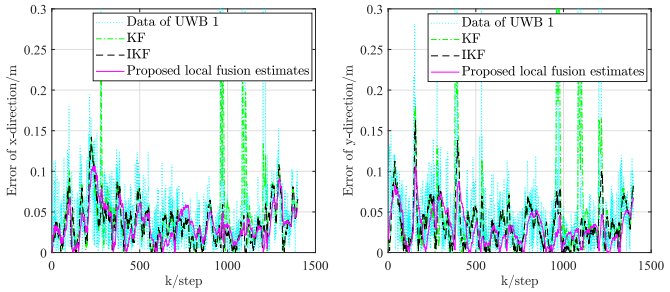


Fig. 4. Error comparisons of UWB 1 (KF considering NLOS identification is indicated by IKF).

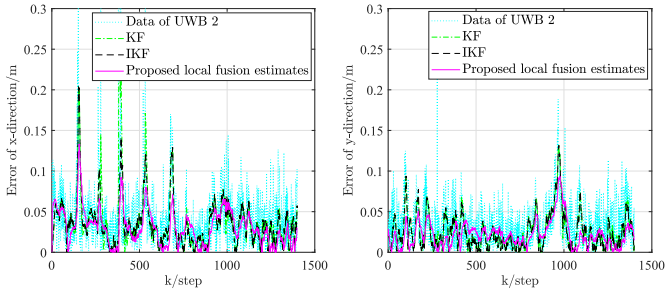


Fig. 5. Error comparisons of UWB 2 (KF considering NLOS identification is indicated by IKF).

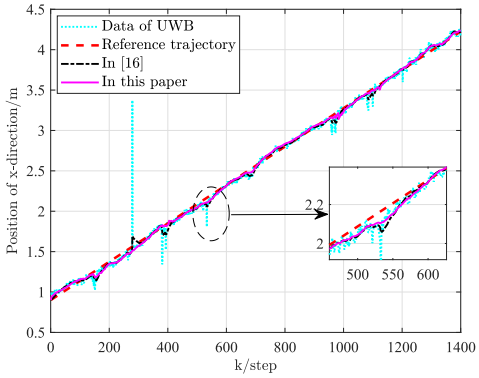


Fig. 6. Position of the target in x-direction.

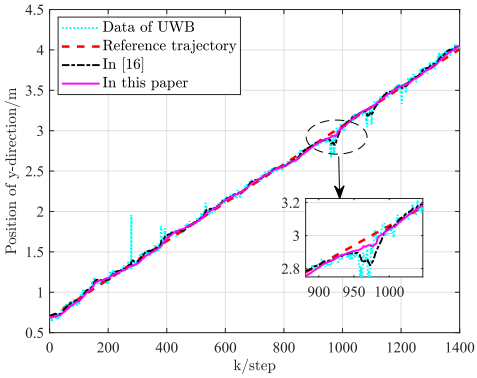


Fig. 7. Position of the target in y-direction.

where $\xi(1,k)$ and $\hat{\xi}(1,k)$ represent the true value of the position along the x_n-axis and its estimated value, respectively. In a similar way, $\xi(2,k)$ and $\hat{\xi}(2,k)$ represent the true value of the position along the y_n-axis and its estimated value, respectively. M denotes the number of sampling. Fig. 8 clearly shows the effectiveness of the proposed method by calculating the RMSE, where it is seen that the RMSE of the modified matrix-weighted fusion estimator is significantly smaller than that of the local modified fusion estimator and the estimator in [16] in general. It is concluded that the modified

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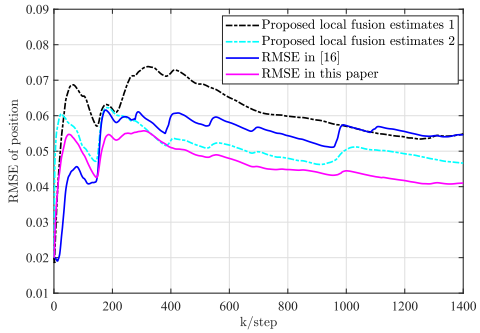


Fig. 8. RMSEs of the target for position.

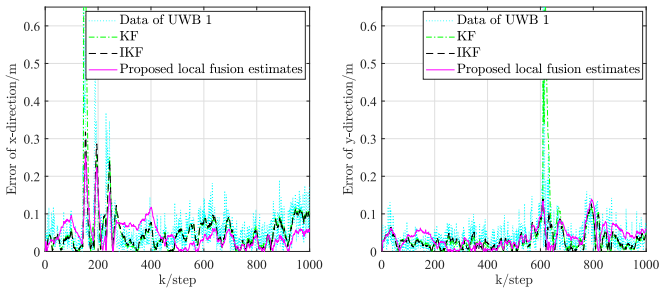


Fig. 9. Error comparisons of UWB 1 (KF considering NLOS identification is indicated by IKF).

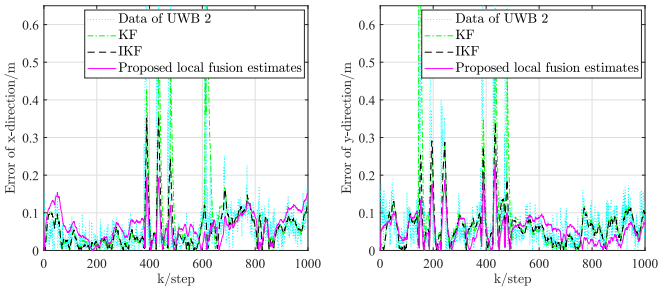


Fig. 10. Error comparisons of UWB 2 (KF considering NLOS identification is indicated by IKF).

matrix-weighted fusion algorithm is more stable and reliable compared with the local modified fusion estimator in the positioning system subject to NLOS environments.

B. Experimental Results With Trajectory as Circle

In this experiment, the target moves along a circle with uniform velocity. Similar to Experiment A, in (27), $\nu_0 = 3$, $\hat{x}_{0|0} = [2.373.290.250.150.54]^{\mathrm{T}}$, $P_{0|0} = \mathrm{diag}[1, 1, 1, 1, 1]$. In (28) and (29), $Q_k = \mathrm{diag}[0.22, 0.22, 10, 10, 1]$, $R_{1,k} = \mathrm{diag}[2.25, 2.25]$, and $R_{2,k} = \mathrm{diag}[2.25, 2.25]$. $\nu_w = 5$, $\nu_i = 3$ and $P_{F,0|0} = \mathrm{diag}[1, 1, 1, 1, 1]$. In addition, $T = 0.02$ s, $\mu = 20$, and $\alpha_i = 1$.

The performance comparisons are given in Figs. 9 and 10. It is shown that the proposed local modified fusion algorithm is more stable than KF considering NLOS identification and KF. In Fig. 10, when $k \in (380, 500)$, it is interfered by obstacles. The errors in x_n-axis direction are over 0.45 m and within 0.36 m by utilizing KF and KF considering NLOS identification,

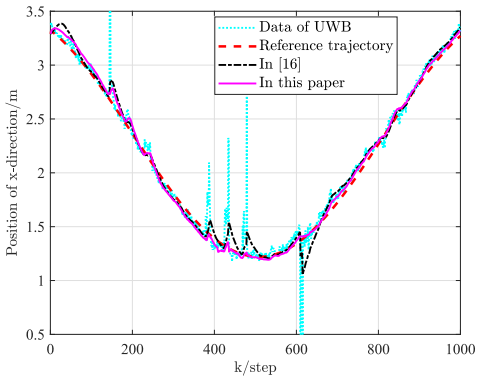


Fig. 11. Position of the target in x-direction.

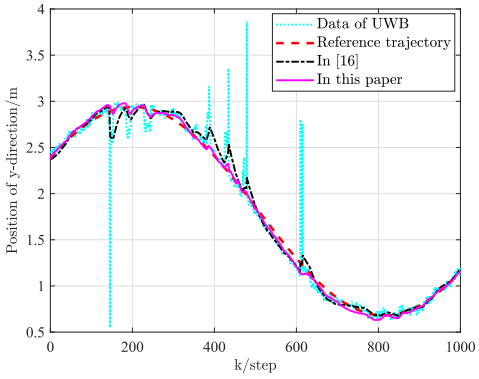


Fig. 12. Position of the target in y-direction.

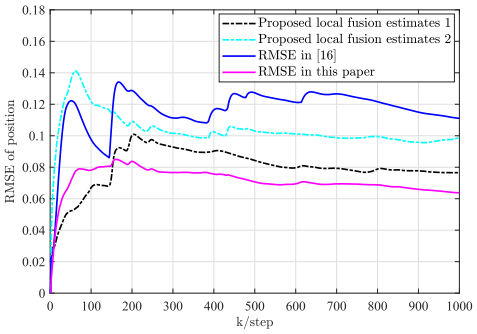


Fig. 13. RMSEs of the target for position.

respectively. However, by using the proposed local modified fusion algorithm, the error is reduced to about 0.2 m.

Figs. 11 and 12 show the position comparisons between the proposed modified matrix-weighted fusion algorithm and the integration algorithm in [16]. It is presented that the results of the proposed Algorithm 1 are closer to the reference trajectory and smoother than that of the integration algorithm in [16] when there are obstacles to interfere. Moreover, from Fig. 13, it is noticed that RMSE of the proposed algorithm is relatively smaller than that of the local modified fusion algorithm in NLOS environments.

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VI. Conclusion

In this article, the problem of modified matrix-weighted fusion estimator has been researched for an indoor positioning system based on IMU and UWB. The measurement of UWB exists outliers caused by NLOS effects. To describe characteristics of non-Gaussian but heavy-tail well, Student's t distribution has been introduced. A local modified fusion estimator has been designed on the basis of a NLOS identification method to fuse the measurements of IMU and UWB, and the fused positioning results have been smoother. In order to obtain accurate estimation, a modified matrix-weighted fusion estimator has been proposed. Sufficient conditions have been given for ensuring exponentially boundedness in mean square for the fusion estimation error. The experimental results illustrate the higher fusion accuracy and smaller uncertainty of the proposed fusion estimator. Future studies will focus on adaptively identifying and excluding measurement outliers caused by multipath and NLOS effects. In addition, it is very important to improve the accuracy of Student's t-modeling for the high-accuracy estimation.

References

[1] D. Feng et al., "An adaptive IMU/UWB fusion method for NLOS indoor positioning and navigation," *IEEE Internet Things J.*, vol. 10, no. 13, pp. 11414–11428, Jul. 2023.

[2] S. Cao, X. Chen, X. Zhang, and X. Chen, "Improving the tracking accuracy of TDMA-based acoustic indoor positioning systems using a novel error correction method," *IEEE Sensors J.*, vol. 22, no. 6, pp. 5427–5436, Mar. 2022.

[3] D.-H. Kim, A. Farhad, and J.-Y. Pyun, "UWB positioning system based on LSTM classification with mitigated NLOS effects," *IEEE Internet Things J.*, vol. 10, no. 2, pp. 1822–1835, Jan. 2023.

[4] R. C. Luo and T.-J. Hsiao, "Indoor localization system based on hybrid wi-Fi/BLE and hierarchical topological fingerprinting approach," *IEEE Trans. Veh. Technol.*, vol. 68, no. 11, pp. 10791–10806, Nov. 2019.

[5] C. Tang, W. Sun, X. Zhang, J. Zheng, W. Wu, and J. Sun, "A novel fingerprint positioning method applying vision-based definition for WIFI-based localization," *IEEE Sensors J.*, vol. 23, no. 14, pp. 16092–16106, Jul. 2023.

[6] X. Xie et al., "Implementation of differential tag sampling for COTS RFID systems," *IEEE Trans. Mobile Comput.*, vol. 19, no. 8, pp. 1848–1861, Aug. 2020.

[7] A. A. N. Shirehjini and S. Shirmohammadi, "Improving accuracy and robustness in HF-RFID-based indoor positioning with Kalman filtering and Tukey smoothing," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 11, pp. 9190–9202, Nov. 2020.

[8] H.-Y. Lin and J.-R. Zhan, "GNSS-denied UAV indoor navigation with UWB incorporated visual inertial odometry," *Measurement*, vol. 206, Jan. 2023, Art. no. 112256.

[9] B. Zhu, X. Tao, J. Zhao, M. Ke, H. Wang, and W. Deng, "An integrated GNSS/UWB/DR/VMM positioning strategy for intelligent vehicles," *IEEE Trans. Veh. Technol.*, vol. 69, no. 10, pp. 10842–10853, Oct. 2020.

[10] D. Gao, X. Zeng, J. Wang, and Y. Su, "Application of LSTM network to improve indoor positioning accuracy," *Sensors*, vol. 20, no. 20, p. 5824, 2020.

[11] C. Jiang, J. Shen, S. Chen, Y. Chen, D. Liu, and Y. Bo, "UWB NLOS/LOS classification using deep learning method," *IEEE Commun. Lett.*, vol. 24, no. 10, pp. 2226–2230, Oct. 2020.

[12] V. Barral, C. J. Escudero, J. A. García-Naya, and R. Maneiro-Catoira, "NLOS identification and mitigation using low-cost UWB devices," *Sensors*, vol. 19, no. 16, p. 3464, 2019.

[13] K. Yu, K. Wen, Y. Li, S. Zhang, and K. Zhang, "A novel NLOS mitigation algorithm for UWB localization in harsh indoor environments," *IEEE Trans. Veh. Technol.*, vol. 68, no. 1, pp. 686–699, Jan. 2019.

[14] S. Bottigliero, D. Milanesio, M. Saccani, and R. Maggiora, "A low-cost indoor real-time locating system based on TDOA estimation of UWB pulse sequences," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, Mar. 2021.

[15] J. Jia, K. Guo, W. Li, X. Yu, and L. Guo, "Composite filtering for UWB-based localization of quadrotor UAV with skewed measurements and uncertain dynamics," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, Feb. 2022.

[16] D. Feng, C. Wang, C. He, Y. Zhuang, and X.-G. Xia, "Kalman-filter-based integration of IMU and UWB for high-accuracy indoor positioning and navigation," *IEEE Internet Things J.*, vol. 7, no. 4, pp. 3133–3146, Apr. 2020.

[17] B. V. Krishnaveni, K. S. Reddy, and P. R. Reddy, "Indoor tracking by adding IMU and UWB using unscented Kalman filter," *Wireless Pers. Commun.*, vol. 123, no. 4, pp. 3575–3596, 2022.

[18] B. Yang, E. Yang, L. Yu, and C. Niu, "Adaptive extended Kalman filter-based fusion approach for high-precision UAV positioning in extremely confined environments," *IEEE/ASME Trans. Mechatronics*, vol. 28, no. 1, pp. 543–554, Feb. 2023.

[19] M. Martalò, S. Perri, G. Verdano, F. De Mola, F. Monica, and G. Ferrari, "Hybrid UWB-inertial TDoA-based target tracking with concentrated anchors," *IEEE Internet Things J.*, vol. 10, no. 14, pp. 12676–12689, Jul. 2023.

[20] Z. Zhi, D. Liu, and L. Liu, "A performance compensation method for GPS/INS integrated navigation system based on CNN–LSTM during GPS outages," *Measurement*, vol. 188, Jan. 2022, Art. no. 110516.

[21] G. Guo, X. Sun, and J. Liu, "5G/GNSS integrated vehicle Localization with adaptive step size Kalman filter," *IEEE Trans. Veh. Technol.*, early access, Jul. 1, 2024, doi: 10.1109/TVT.2024.3421383.

[22] J. Wang, T. Zhang, B. Jin, Y. Zhu, and J. Tong, "Student's t-based robust Kalman filter for a SINS/USBL integration navigation strategy," *IEEE Sensors J.*, vol. 20, no. 10, pp. 5540–5553, May 2020.

[23] Y. Huang, Y. Zhang, N. Li, and J. Chambers, "Robust student's t-based nonlinear filter and smoother," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 52, no. 5, pp. 2586–2596, Oct. 2016.

[24] L. Yan, C. Di, Q. J. Wu, Y. Xia, and S. Liu, "Distributed fusion estimation for multisensor systems with non-Gaussian but heavy-tailed noises," *ISA Trans.*, vol. 101, pp. 160–169, Jun. 2020.

[25] L. Zhao, X. Cao, L. Li, and H. Yang, "Event-triggered distributed fusion for multirate multisensor systems with heavy-tailed noises," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 5, pp. 3137–3150, May 2022.

[26] H. Li, L. Yan, Y. Zhou, Y. Xia, and X. Shi, "Sequential fusion estimation for Markov jump systems with heavy-tailed noises," *Int. J. Syst. Sci.*, vol. 54, no. 9, pp. 1910–1925, 2023.

[27] L. Li and Y. Xia, "Stochastic stability of the unscented Kalman filter with intermittent observations," *Automatica*, vol. 48, no. 5, pp. 978–981, 2012.

[28] Y. Liu, H. Zhi, J. Wei, X. Zhu, and Q. Zhu, "Event-triggered control for linear continuous switched singular systems," *Appl. Math. Comput.*, vol. 374, Jun. 2020, Art. no. 125038.

[29] L. Xie and P. P. Khargonekar, "Lyapunov-based adaptive state estimation for a class of nonlinear stochastic systems," *Automatica*, vol. 48, no. 7, pp. 1423–1431, 2012.

[30] S.-L. Sun, "Multi-sensor optimal information fusion Kalman filters with applications," *Aerosp. Sci. Technol.*, vol. 8, no. 1, pp. 57–62, 2004.

[31] R. Wang and S. Sun, "Distributed matrix-weighted fusion consensus filtering with two-stage filtering for sensor networks," *IEEE Sensors J.*, vol. 23, no. 5, pp. 5003–5013, Mar. 2023.

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